Sampling Techniques

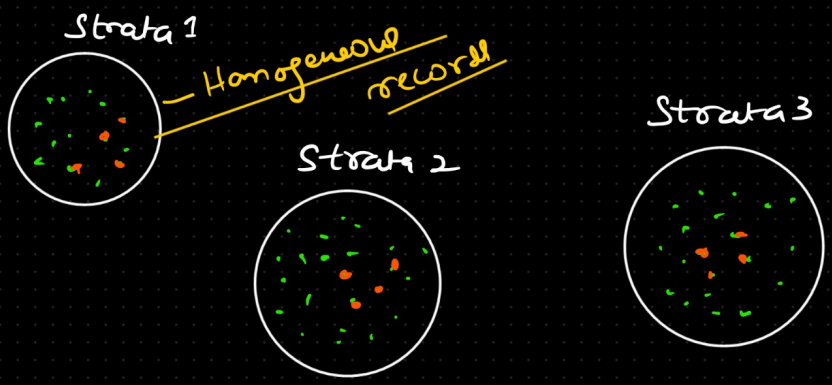
① Random Sampling ⟶

fetch the records randomly

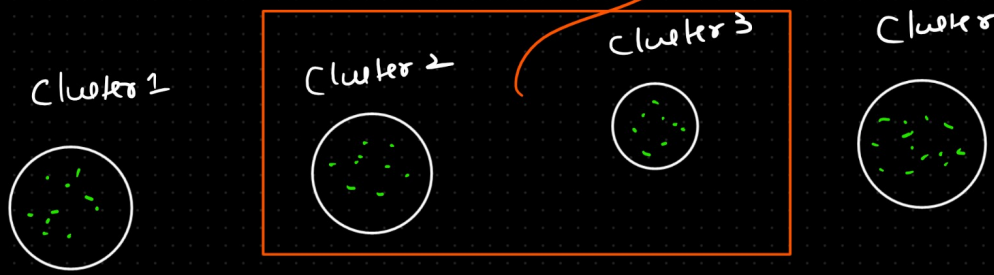


Each case is equally likely to be selected.

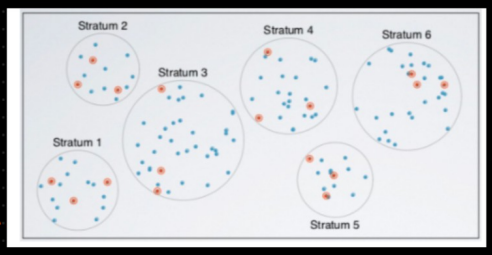
② Stratified Sampling ⟶ to get rid of the biasness in the given model to get the accurate results



③ Cluster Sampling



Randomly →

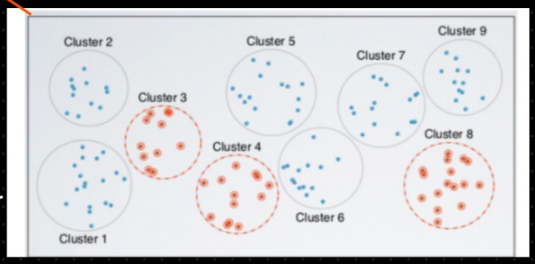


→ Stratified Sampling

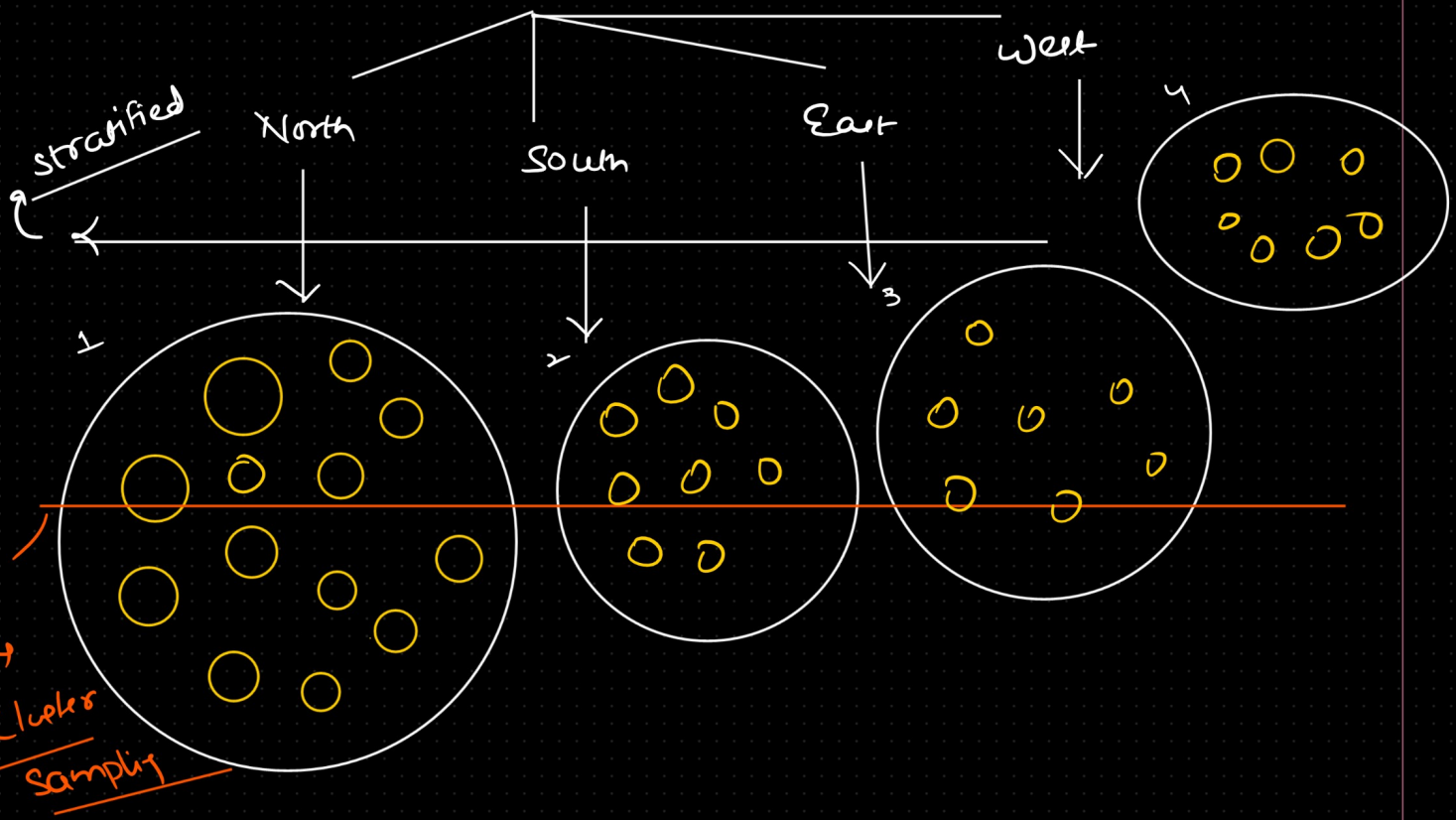
↳ Diagnostic Classification Problem

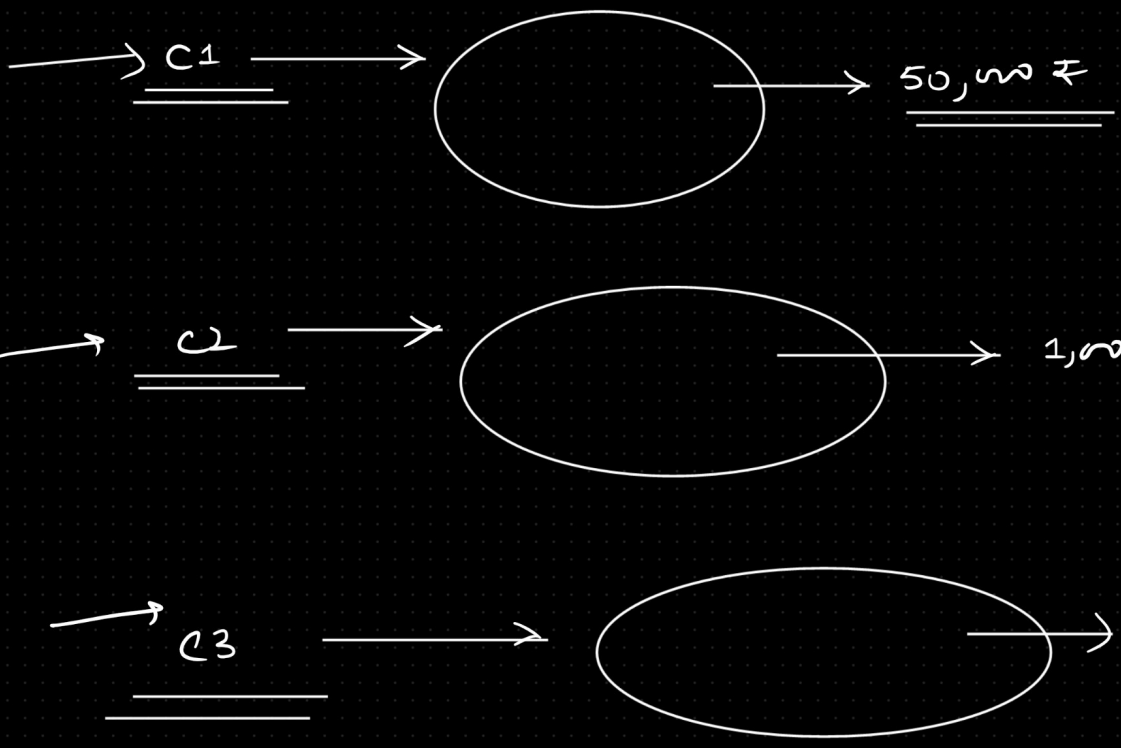
Group represents similar set of items

Randomly

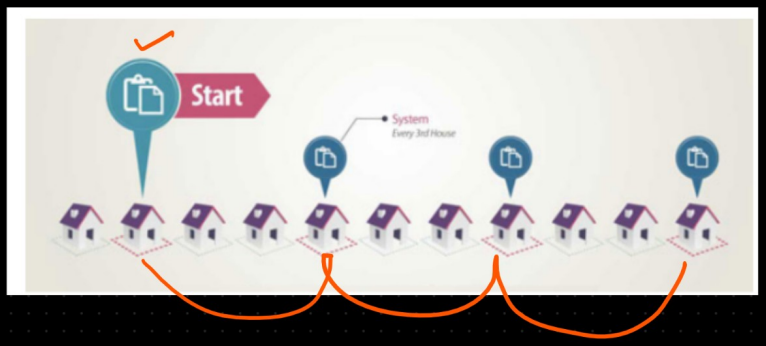


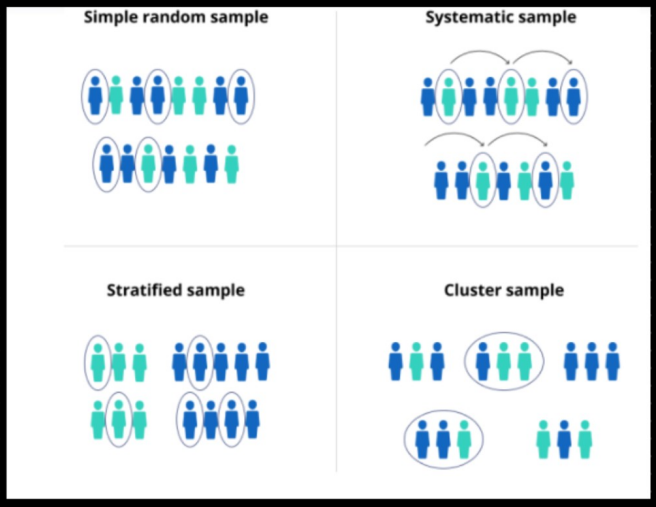
→ Cluster Sampling





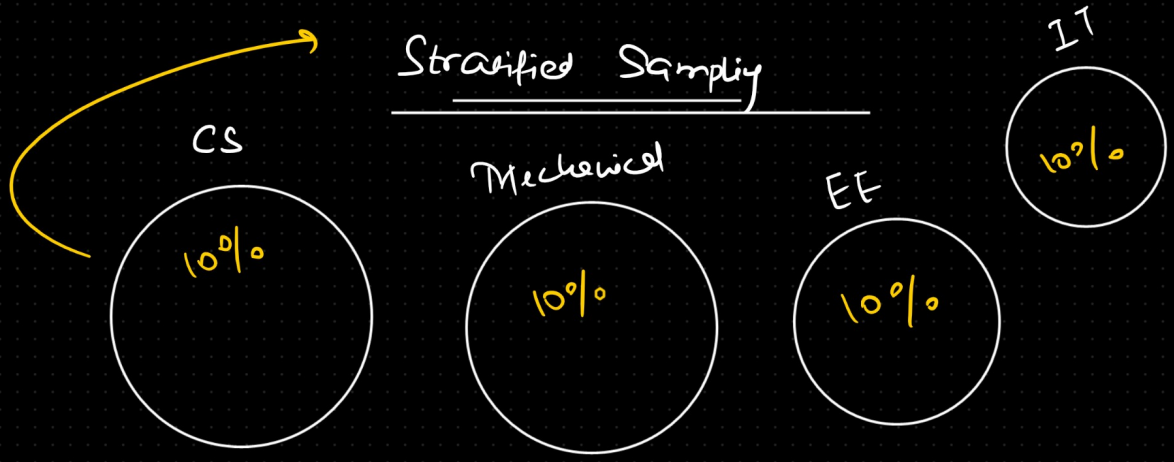
Systematic Samplig





Screening

A university wants to understand the average GPA of its students. It randomly selects 10% of students from each department (e.g., CS, Mechanical, EE, etc.) to ensure all departments are represented.



A retail chain wants feedback from its customers. It chooses 5 cities randomly and then surveys all customers visiting their stores in those cities.

→ Cluster Sampling

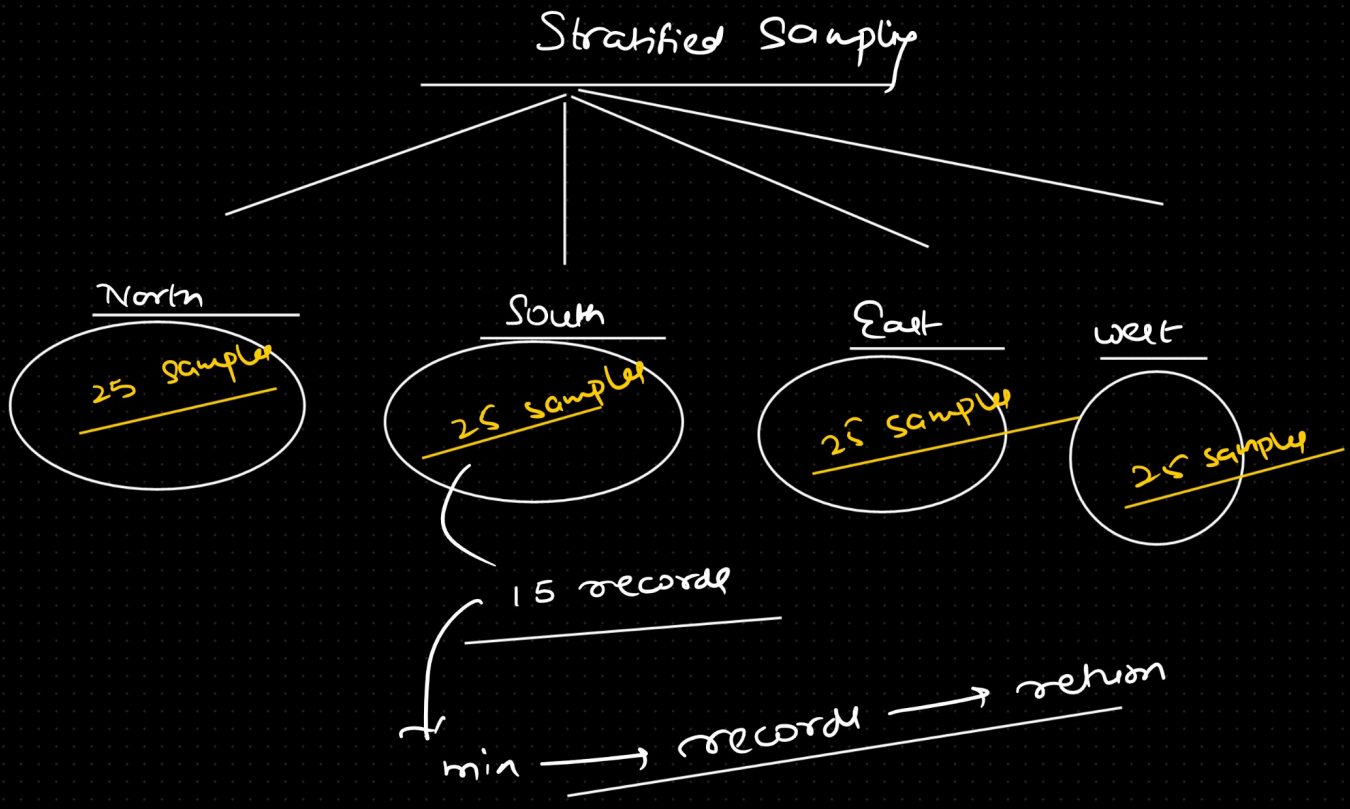
Kth item

A factory quality control team checks every 10th item coming off a production line to look for defects.

→ Systematic sampling

You want to know the food preferences of people in a city. You randomly pick 1,000 residents from the city's database and survey them.

→ Simple Random Sampling


Stratified Sampling
North
25 samples
South
25 samples
East
25 samples
West
25 samples
15 records
min → records → return